

Nicolò Bonacorsi

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Education

Columbia University

MS in Applied Mathematics

- Teaching Assistant – APMA E4300 Numerical Methods.

Sapienza University of Rome

BS in Mathematical Sciences for Artificial Intelligence, GPA: 3.99/4.00

Final Grade: 110/110 cum laude

University of Bologna

BS in Statistical Sciences, GPA: 3.95/4.00

Final Grade: 110/110 cum laude; Award: University of Bologna Prize for Outstanding Student

New York, NY

Aug 2025 – Dec 2026

Jan 2026 – Present

Rome, Italy

Sep 2022 – Sep 2025

Bologna, Italy

Sep 2021 – Jul 2024

Publications

- Nicolò Bonacorsi, Michele Caprio, Francesca Meneghello, and Francesco Restuccia (2026). *Credal Rate-Distortion Theory*. To appear in the proceedings of the 2026 IEEE Information Theory Workshop (ITW). *Mentis Laboratory preprint*.
- Nicolò Bonacorsi and Matteo Bordoni (2026). *Bayesian Modeling of Collatz Stopping Times: A Probabilistic Machine Learning Perspective*. arXiv:2603.04479. <https://arxiv.org/abs/2603.04479>.

Research Experience

Columbia University Supervisor: Professor Ioannis Karatzas

Graduate Researcher

- Reduced a diffusion-control problem with jointly optimized drift, volatility, and stopping to a one-dimensional free-boundary system, yielding a tractable characterization of optimal state-dependent feedback controls and stopping thresholds.
- Established value-function envelope, regularity, and verification results for finite-cost admissible controls, including degenerate zero-volatility policies not covered by classical smooth-verification arguments.
- Classified when optimal policies collapse to closed-form constant-control thresholds versus genuinely state-dependent controls, recovering the Beneš-Gaitsgory-Karatzas model as a special constant-curvature regime.

Northeastern University Supervisor: Professor Francesco Restuccia

Graduate Researcher

- Formulated a rate-distortion framework for lossy compression under set-valued predictive uncertainty, enabling compressed representations to preserve ambiguity relevant to downstream decisions rather than replace it with a single estimated law.
- Proved that zero-distortion credal coding reduces to graph coloring, converting minimum codebook size and output entropy into chromatic optimization problems; derived compression laws for local, finite-window, and unrestricted observers.
- Implemented an LP and graph-coloring evaluation pipeline that solved 54,000 set-membership programs and compressed 60 SVHN predictive credal sets into 14 codewords ($4.29\times$) with no incompatibility violations at the prescribed tolerance, versus 18.6% incompatible within-cluster pairs for an equal-size barycentric baseline.

Projects

Independent Quantitative Research

Bootstrap-Calibrated Distributionally Robust Allocation

- Constructed a distributionally robust allocation pipeline in Python for 25 Fama-French size-value portfolios, calibrating a covariance ambiguity set via circular-block bootstrap resampling and solving capped long-only quadratic programs with a custom active-set solver.
- Achieved an 18.4% reduction in annualized volatility versus equal weight over the 2013–2026 walk-forward period after model selection on 2000–2012 data only (16.45% vs. 20.15%; paired block-bootstrap 95% CI: 15.0%–22.6%); delivered a 12.90% CAGR and 0.82 net Sharpe after 10 bps costs, matching validation-selected Ledoit-Wolf GMV at 16.46%.

Independent Quantitative Research

Maturity-Safe Distributional Volatility Forecasting

- Combined six VIX, HAR-style, and EWMA variance models in Python using bounded-loss Hedge updates applied only after 22-day targets matured; calibrated a 90% upper variance forecast from 22 interlaced residual streams with non-overlapping targets within each stream.
- Improved out-of-sample volatility forecasting over 2013–2026, lowering log-RMSE by 14.9% versus VIX-implied variance (HAC-robust $p < 0.001$) and achieving 89% coverage at a nominal 90% level across 3,381 forecasts; cost-aware SPY controller delivered 8.3% annualized volatility, 0.84 Sharpe ratio over cash, and 11.6% maximum drawdown after 1 bp costs.

Columbia University

Bayesian Modeling of Collatz Stopping Times

- Modeled Collatz stopping times over integers up to 10 million using a scalable hierarchical Bayesian negative-binomial specification with partial pooling across modulo-8 residue classes, capturing overdispersion and right-tail heterogeneity.
- Calibrated a Monte Carlo simulation engine via Bayesian updating; regime conditioning reduced Wasserstein distance to the empirical distribution by $3.2\times$, improving tail-risk representation, posterior calibration, and distributional fit.

Professional Experience

EndParty Group

Co-Founder, Quantitative Strategy & Analytics

- Designed forecasting models for event revenues and sponsorship returns from historical demand, behavioral indicators, and macro proxies; used outputs to guide capital-allocation and event-planning decisions across 4 years of operations.
- Converted sponsor-ROI analyses into data-backed commercial proposals that helped secure 12 partnerships and contributed to a $12\times$ increase in company revenue from the 2021 baseline to 2023.

Skills

Programming & Data: Python, SQL, R, MATLAB; NumPy, pandas, SciPy, scikit-learn, PyTorch, statsmodels.

Statistical & ML Methods: Time Series, Bayesian Inference, Monte Carlo, GLMs, Statistical Learning, Neural Networks, Model Validation, Risk Modeling.

Mathematics: Stochastic Processes, Stochastic Control, Optimal Stopping, HJB Equations, Optimization, Numerical Methods, PCA/SVD, Information Theory, Uncertainty Quantification.

Languages: Italian (native), English (fluent).